

Expectation

1. Idea of expectation

A random variable (RV) takes a real value each time a random experiment is performed.

Expectation seeks to represent the average of these values over a large number of experiments.

2. Motivation (Discrete Cases)

Let:

- X = discrete RV
- pmf p_X
- support E_X - This is the support of X . The set of all possible values that X can take.
- $\mathbb{E}[X]$ - Expectation (expected value) of the random variable X
 - $p_X(k)$ - This is the probability mass function (PMF).

Suppose we perform **N experiments**.

Let:

$$N_k = \# \text{ times } X = k$$

Then empirically:

$$\frac{N_k}{N} \rightarrow p_X(k)$$

The sample average:

$$\frac{1}{N} \sum_{k \in E_X} k N_k = \sum_{k \in E_X} k \frac{N_k}{N}$$

Taking limit as $N \rightarrow \infty$:

$$\mathbb{E}[X] = \sum_{k \in E_X} k p_X(k)$$

k is every possible value k can take while performing N experiments

E_X -

Interpretation: Weighted average of values X takes

3. Definition

DISCRETE RV

If X is discrete with pmf p_X :

$$\mathbb{E}[X] = \sum_{k \in E_X} k p_X(k)$$

CONTINUOUS RV

If X is continuous with pdf $f_X(x)$:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

This is the **general definition of expectation**.

4. Examples

1. UNIFORM DISTRIBUTION

$$U \sim \text{Unif}[a, b]$$

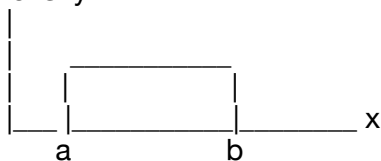
PDF:

$$f_U(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

Expectation:

$$\mathbb{E}[U] = \int_a^b x \frac{1}{b-a} dx = \frac{a+b}{2}$$

Density



Mean lies exactly at midpoint:

$$(a+b)/2$$

2. EXPONENTIAL DISTRIBUTION

$$X \sim \text{Exp}(\mu)$$

PDF:

$$f_X(x) = \mu e^{-\mu x}, \quad x \geq 0$$

$$\mathbb{E}[X] = \int_0^{\infty} x \mu e^{-\mu x} dx$$

Substitute:

$$t = \mu x$$

Then:

$$\mathbb{E}[X] = \frac{1}{\mu}$$

Note:

$$\int_0^{\infty} y e^{-y} dy = 1$$

(Gamma distribution with parameter 1)

3. BINOMIAL DISTRIBUTION

$$N \sim \text{Bin}(n, p)$$

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mathbb{E}[N] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$$

Using:

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

Differentiate:

$$n(1+x)^{n-1} = \sum_{k=0}^n k \binom{n}{k} x^{k-1}$$

After algebra:

$$\mathbb{E}[N] = np$$

4. NORMAL DISTRIBUTION

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x e^{-(x-\mu)^2/(2\sigma^2)} dx$$

Substitute:

$$y = x - \mu$$

Then:

$$\mathbb{E}[X] = \mu + \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} y e^{-y^2/(2\sigma^2)} dy$$

Since integrand is odd function over symmetric limits:

$$\int_{-\infty}^{\infty} y e^{-y^2/(2\sigma^2)} dy = 0$$

Thus:

$$\mathbb{E}[X] = \mu$$

ADDITIONAL RESULTS

$$N \sim \text{Geo}(p)$$

$$p_X(k) = (1-p)^{k-1}p, \quad k \geq 1$$

$$\mathbb{E}[X] = \frac{1}{p}$$

$$N \sim \text{Poi}(\lambda)$$

$$\mathbb{E}[N] = \lambda$$

5. CAUCHY DISTRIBUTION

$$X \sim \text{Cauchy RV}$$

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx$$

Splitting:

$$\int_0^{\infty} \frac{x}{\pi(1+x^2)} dx + \int_{-\infty}^0 \frac{x}{\pi(1+x^2)} dx$$

Both integrals diverge:

$$= \infty - \infty$$

Expectation is undefined

6. PARETO DISTRIBUTION

$$X \sim \text{Pareto}(x_0, \alpha)$$

where:

$$x_0, \alpha > 0$$

Tail probability:

$$P(X > x) = \begin{cases} \left(\frac{x}{x_0}\right)^{-\alpha} & x \geq x_0 \\ 1 & x < x_0 \end{cases}$$

PDF:

$$f_X(x) = \begin{cases} \frac{\alpha}{x_0^\alpha} x^{-(\alpha+1)} & x \geq x_0 \\ 0 & \text{otherwise} \end{cases}$$

Expectation:

$$\mathbb{E}[X] = \int_{x_0}^{\infty} \alpha x_0^\alpha x^{-(\alpha+1)} x dx$$

$$= \frac{\alpha}{\alpha - 1} x_0 \quad \text{if } \alpha > 1$$

$$= \infty \quad \text{if } \alpha \leq 1$$

Interpretation:

Infinite expectation for $\alpha \leq 1$ because tail probabilities decay polynomially, not exponentially.

Key Conceptual Takeaways

1. Expectation = long-run average
2. Discrete $\rightarrow$ summation
3. Continuous $\rightarrow$ integral
4. Symmetry can simplify integrals (Normal case)

5. Heavy-tailed distributions may have:
- Infinite mean (Pareto)
 - Undefined mean (Cauchy)